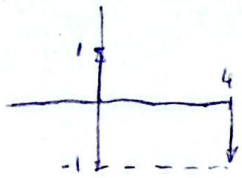
TS1

$h[n] = \delta[n] - \delta[n-4]$ $\qquad$ $Y[n] = X[n] * h[n]$

a) $x[n] = \cos(W_{on} \cdot T_s)$ $\qquad$ Expresar la respuesta como: $A\cos(W_o \cdot n \cdot T_s + \phi)$

$$Y[n] = \cos(W_{on}T_s) * (\delta[n] - \delta[n-4])$$



$$Y[n] = \cos(W_{on}T_s) * \delta[n] - \underbrace{\cos(W_oT_sn) * \delta[n-4]}$$

$$Y[n] = \cos(W_{on}T_s) - \cos(W_oT_s(n-4))$$

$$Y[n] = \cos(W_{on}T_s) - \cos(W_oT_sn - W_oT_s4)$$

Aplico que $\cos(A \pm B) = \cos(A)\cos(B) \mp \operatorname{sen}(A)\operatorname{sen}(B)$

$$= \cos(W_{on}T_s) - \cos(W_oT_sn)\cos(W_oT_s4) - \operatorname{sen}(W_oT_sn)\operatorname{sen}(W_oT_s4)$$

$$= (1 - \cos(W_oT_s4))\cos(W_{on}T_s) - \operatorname{sen}(W_oT_sn)\operatorname{sen}(W_oT_s4) = A\cos(W_{on}T_s + \phi)$$

$$\Rightarrow = A\cos(W_{on}T_s)\cos(\phi) - A\operatorname{sen}(W_{on}T_s)\operatorname{sen}(\phi)$$

$$\Rightarrow A\cos(\phi) = 1 - \cos(W_oT_s4)$$
$$A\operatorname{sen}(\phi) = \operatorname{sen}(W_oT_s4)$$

elevo Ambos terminos al cuadrado

$$(A\cos(\phi))^2 = (1 - \cos(W_oT_s4))^2$$
$$(A\operatorname{sen}(\phi))^2 = (\operatorname{sen}(W_oT_s4))^2$$

$$\Rightarrow (1)\ A^2\cos^2(\phi) = 1 - 2\cos(W_oT_s4) + \cos^2(W_oT_s4)$$
$$+\ (2)\ A^2\operatorname{sen}^2(\phi) = \operatorname{sen}^2(W_oT_s4)$$

Los sumo

$$A^2(\underbrace{\cos^2(\phi) + \operatorname{sen}^2(\phi)}_{1}) = 1 - 2\cos(W_oT_s4) + \overbrace{\cos^2(W_oT_s4) + \operatorname{sen}^2(W_oT_s4)}^{1}$$

$$\Rightarrow A^2 = 1 - 2\cos(W_oT_s4) + 1 \Rightarrow \boxed{A = \sqrt{2 - 2\cos(W_oT_s4)}}$$

(1) $A\cos(\phi) = 1 - \cos(W_0 T_s 4)$

(2) $A\,\text{sen}(\phi) = \text{sen}(W_0 T_s 4)$

hago (2)/(1)

$$\Rightarrow \frac{\cancel{A}\,\text{sen}(\phi)}{\cancel{A}\cos(\phi)} = \frac{\text{sen}(W_0 T_s 4)}{1-\cos(W_0 T_s 4)} \Rightarrow \text{tag}(\phi) = \frac{\text{sen}(W_0 T_s 4)}{1-\cos(W_0 T_s 4)}$$

$$\Rightarrow \boxed{\phi = \text{tg}^{-1}\left(\frac{\text{sen}(W_0 T_s 4)}{1-\cos(W_0 T_s 4)}\right)}$$

$$\Rightarrow Y[n] = \sqrt{2-2\cos(W_0 T_s 4)}\,\cos\left(W_0 n T_s + \text{tg}^{-1}\left(\frac{\text{sen}(W_0 T_s 4)}{1-\cos(W_0 T_s 4)}\right)\right)$$

b) $X[n] = \left(\frac{1}{2}\right)^n u[n]$

$$Y[n] = \left(\tfrac{1}{2}\right)^n u[n] * (\delta[n] - \delta[n-4])$$

$$Y[n] = \left(\tfrac{1}{2}\right)^n u[n] * \delta[n] - \left(\tfrac{1}{2}\right)^n u[n] * \delta[n-4]$$

$$Y[n] = \left(\tfrac{1}{2}\right)^n u[n] - \left(\tfrac{1}{2}\right)^{n-4} u[n-4]$$

$$Y[n] = \left(\tfrac{1}{2}\right)^n \left(u[n] - \frac{1}{\left(\frac{1}{2}\right)^4} u[n-4]\right) = \left(\tfrac{1}{2}\right)^n \left(u[n] - 2^4 u[n-4]\right)$$

c) $X[n] = u[n+1] - u[n-2]$

$$Y[n] = (u[n+1] - u[n-2]) * (\delta[n] - \delta[n-4])$$

$$Y[n] = u[n+1] * \delta[n] - u[n+1] * \delta[n-4] - u[n-2] * \delta[n] + u[n-2] * \delta[n-4]$$

$$Y[n] = u[n+1] - u[n-3] - u[n-2] + u[n-6]$$

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